(which is a restriction of the signum function) clearly fails to satisfy the intermediate value property on the interval $[-1, 1]$. Therefore, by Darboux's Theorem, there does not exist a function f such that $f'(x) = g(x)$ for all $x \in [-1, 1]$. In other words, g is *not* the derivative on $[-1, 1]$ of any function. □

Exercises for Section 6.2

1. For each of the following functions on $\mathbb{R}$ to $\mathbb{R}$, find points of relative extrema, the intervals on which the function is increasing, and those on which it is decreasing:
   (a) $f(x) := x^2 - 3x + 5$, (b) $g(x) := 3x - 4x^2$,
   (c) $h(x) := x^3 - 3x - 4$, (d) $k(x) := x^4 + 2x^2 - 4$.

2. Find the points of relative extrema, the intervals on which the following functions are increasing, and those on which they are decreasing:
   (a) $f(x) := x + 1/x$ for $x \neq 0$, (b) $g(x) := x/(x^2 + 1)$ for $x \in \mathbb{R}$,
   (c) $h(x) := \sqrt{x} - 2\sqrt{x+2}$ for $x > 0$, (d) $k(x) := 2x + 1/x^2$ for $x \neq 0$.

3. Find the points of relative extrema of the following functions on the specified domain:
   (a) $f(x) := |x^2 - 1|$ for $-4 \leq x \leq 4$, (b) $g(x) := 1 - (x-1)^{2/3}$ for $0 \leq x \leq 2$,
   (c) $h(x) := x|x^2 - 12|$ for $-2 \leq x \leq 3$, (d) $k(x) := x(x-8)^{1/3}$ for $0 \leq x \leq 9$.

4. Let $a_1, a_2, \ldots, a_n$ be real numbers and let f be defined on $\mathbb{R}$ by
$$f(x) := \sum_{i=1}^{n} (a_i - x)^2 \quad \text{for} \quad x \in \mathbb{R}.$$
   Find the unique point of relative minimum for f.

5. Let $a > b > 0$ and let $n \in \mathbb{N}$ satisfy $n \geq 2$. Prove that $a^{1/n} - b^{1/n} < (a-b)^{1/n}$. [*Hint*: Show that $f(x) := x^{1/n} - (x-1)^{1/n}$ is decreasing for $x \geq 1$, and evaluate f at 1 and a/b.]

6. Use the Mean Value Theorem to prove that $|\sin x - \sin y| \leq |x - y|$ for all x, y in $\mathbb{R}$.

7. Use the Mean Value Theorem to prove that $(x-1)/x < \ln x < x - 1$ for $x > 1$. [*Hint*: Use the fact that $D \ln x = 1/x$ for $x > 0$.]

8. Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable in (a, b). Show that if $\lim_{x \to a} f'(x) = A$, then $f'(a)$ exists and equals A. [*Hint*: Use the definition of $f'(a)$ and the Mean Value Theorem.]

9. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) := 2x^4 + x^4 \sin(1/x)$ for $x \neq 0$ and $f(0) := 0$. Show that f has an absolute minimum at $x = 0$, but that its derivative has both positive and negative values in every neighborhood of 0.

10. Let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) := x + 2x^2 \sin(1/x)$ for $x \neq 0$ and $g(0) := 0$. Show that $g'(0) = 1$, but in every neighborhood of 0 the derivative $g'(x)$ takes on both positive and negative values. Thus g is not monotonic in any neighborhood of 0.

11. Give an example of a uniformly continuous function on [0,1] that is differentiable on (0, 1) but whose derivative is not bounded on (0, 1).

12. If $h(x) := 0$ for $x < 0$ and $h(x) := 1$ for $x \geq 0$, prove there does not exist a function $f : \mathbb{R} \to \mathbb{R}$ such that $f'(x) = h(x)$ for all $x \in \mathbb{R}$. Give examples of two functions, not differing by a constant, whose derivatives equal $h(x)$ for all $x \neq 0$.

13. Let I be an interval and let $f : I \to \mathbb{R}$ be differentiable on I. Show that if f' is positive on I, then f is strictly increasing on I.

14. Let I be an interval and let $f : I \to \mathbb{R}$ be differentiable on I. Show that if the derivative f' is never 0 on I, then either $f'(x) > 0$ for all $x \in I$ or $f'(x) < 0$ for all $x \in I$.

15. Let I be an interval. Prove that if f is differentiable on I and if the derivative f' is bounded on I, then f satisfies a Lipschitz condition on I. (See Definition 5.4.4.)

16. Let $f : [0, \infty) \to \mathbb{R}$ be differentiable on $(0, \infty)$ and assume that $f'(x) \to b$ as $x \to \infty$.
    (a) Show that for any $h > 0$, we have $\lim_{x\to\infty} (f(x+h) - f(x))/h = b$.
    (b) Show that if $f(x) \to a$ as $x \to \infty$, then $b = 0$.
    (c) Show that $\lim_{x\to\infty} (f(x)/x) = b$.

17. Let f, g be differentiable on $\mathbb{R}$ and suppose that $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$. Show that $f(x) \leq g(x)$ for all $x \geq 0$.

18. Let $I := [a, b]$ and let $f : I \to \mathbb{R}$ be differentiable at $c \in I$. Show that for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $0 < |x - y| < \delta$ and $a \leq x \leq c \leq y \leq b$, then
$$\left| \frac{f(x) - f(y)}{x - y} - f'(c) \right| < \varepsilon.$$

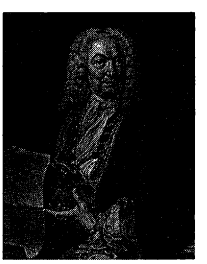
19. A differentiable function $f : I \to \mathbb{R}$ is said to be **uniformly differentiable** on $I := [a, b]$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $0 < |x - y| < \delta$ and $x, y \in I$, then
$$\left| \frac{f(x) - f(y)}{x - y} - f'(x) \right| < \varepsilon.$$
Show that if f is uniformly differentiable on I, then f' is continuous on I.

20. Suppose that $f : [0, 2] \to \mathbb{R}$ is continuous on $[0, 2]$ and differentiable on $(0, 2)$, and that $f(0) = 0, f(1) = 1, f(2) = 1$.
    (a) Show that there exists $c_1 \in (0, 1)$ such that $f'(c_1) = 1$.
    (b) Show that there exists $c_2 \in (1, 2)$ such that $f'(c_2) = 0$.
    (c) Show that there exists $c \in (0, 2)$ such that $f'(c) = 1/3$.

Section 6.3 L'Hospital's Rules

In this section we will discuss limit theorems that involve cases that cannot be determined by previous limit theorems. For example, if $f(x)$ and $g(x)$ both approach 0 as x approaches a, then the quotient $f(x)/g(x)$ may or may not have a limit at a and it is said to have the indeterminate form 0/0. The limit theorem for this case is due to Johann Bernoulli and first appeared in the 1696 book published by L'Hospital.

Johann Bernoulli



Johann Bernoulli (1667–1748) was born in Basel, Switzerland. Johann worked for a year in his father's spice business, but he was not a success. He enrolled in Basel University to study medicine, but his brother Jacob, twelve years older and a Professor of Mathematics, led him into mathematics. Together, they studied the papers of Leibniz on the new subject of calculus. Johann received his doctorate at Basel University and joined the faculty at Groningen in Holland, but upon Jacob's death in 1705, he returned to Basel and was awarded Jacob's chair in mathematics. Because of his many advances in the subject, Johann is regarded as one of the founders of calculus.

While in Paris in 1692, Johann met the Marquis Guillame Francois de L'Hospital and agreed to a financial arrangement under which he would teach the new calculus to L'Hospital, giving L'Hospital the right to use Bernoulli's lessons as he pleased. This was subsequently continued through a series of letters. In 1696, the first book on differential calculus, *L'Analyse des Infiniment Petits*, was published by L'Hospital. Though L'Hospital's name was not on the title page, his portrait was on the frontispiece and the preface states "I am indebted to the clarifications of the brothers Bernoulli, especially the younger." The book contains a theorem on limits later known as L'Hospital's Rule although it was in fact discovered by Johann Bernoulli. In 1922, manuscripts were discovered that confirmed the book consisted mainly of Bernoulli's lessons. And in 1955, the L'Hospital–Bernoulli correspondence was published in Germany.